

Chevalley–Warning divisibility for conjecture 00000009496

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Statement and interpretation

The official conjecture describes Chevalley–Warning divisibility for a system of polynomial equations, with the degrees combined by their sum. The precise standard assertion is the following. Here all degrees are total degrees, and the empty family is allowed.

Theorem 1. *Let p be prime, let $n, r \geq 0$, and let $f_1, \dots, f_r \in \mathbb{F}_p[X_1, \dots, X_n]$. If*

$$\sum_{i=1}^r \deg f_i < n,$$

then the number

$$N = \#\{x \in \mathbb{F}_p^n : f_i(x) = 0 \text{ for every } 1 \leq i \leq r\}$$

is divisible by p .

The strict inequality is the standard Chevalley–Warning hypothesis. The conjecture gives no separate numerical meaning to its phrase about the “combination” of divisibility; this report uses the degree-sum condition above and makes no stronger claim.

Elementary proof

Lemma 1. *For each integer e with $0 \leq e < p - 1$, $\sum_{t \in \mathbb{F}_p} t^e = 0$ in $\mathbb{F}_p$.*

Proof. When $e = 0$, every summand is 1, including 0^0 as used in polynomial evaluation, so the sum is $p = 0$ in $\mathbb{F}_p$. For $1 \leq e < p - 1$, choose a generator g of the cyclic group $\mathbb{F}_p^\times$. Then $g^e \neq 1$ and

$$(g^e - 1) \sum_{t \in \mathbb{F}_p} t^e = (g^e - 1) \sum_{k=0}^{p-2} g^{ke} = g^{e(p-1)} - 1 = 0.$$

Since $g^e - 1 \neq 0$ in a field, the sum is zero. □

Proof of Theorem 1. Put $q = p - 1$ and form the polynomial

$$P(X) = \prod_{i=1}^r (1 - f_i(X)^q).$$

For each $x \in \mathbb{F}_p^n$, Fermat’s theorem gives $a^q = 1$ for $a \neq 0$, while $0^q = 0$ because $q > 0$. Thus $P(x)$ is exactly the indicator, in $\mathbb{F}_p$, of the common zero set. Consequently

$$\sum_{x \in \mathbb{F}_p^n} P(x) = N \cdot 1_{\mathbb{F}_p}.$$

Moreover,

$$\deg P \leq q \sum_{i=1}^r \deg f_i < qn.$$

Each monomial $cX_1^{e_1} \cdots X_n^{e_n}$ occurring in P therefore has $\sum_j e_j < qn$. At least one exponent, say e_j , is less than q ; otherwise every exponent would be at least q and their sum at least qn . Summation over all $x \in \mathbb{F}_p^n$ factors as

$$\sum_x c \prod_{j=1}^n x_j^{e_j} = c \prod_{j=1}^n \left(\sum_{t \in \mathbb{F}_p} t^{e_j} \right) = 0$$

by Lemma 1. Summing the monomials of P gives $N \cdot 1_{\mathbb{F}_p} = 0$. The kernel of the canonical map $\mathbb{Z} \rightarrow \mathbb{F}_p$ is $p\mathbb{Z}$, so $p \mid N$. $\square$

Correspondence with the Lean theorem

In `lean/Main.lean`, the theorem is stated for an arbitrary finite field K , a finite variable index type σ , and a finite equation index type ι . For the prime-field assertion take $K = \mathbb{F}_p$ (represented by `ZMod p`), $\sigma = \text{Fin } n$, and $\iota = \text{Fin } r$. Its family f consists of actual multivariate polynomials. The hypothesis sums their actual `totalDegree` values. The conclusion counts the subtype of coordinate vectors satisfying every polynomial equation. Mathlib's theorem `char_dvd_card_solutions_of_fintype_sum_lt` proves exactly this finite-field assertion; the submitted final theorem specializes its field to an arbitrary finite field, which includes `ZMod p` for prime p . The imported theorem contains the indicator-polynomial argument above. Both the final theorem and this dependency are audited with `#print axioms`.

Reproduction

The Lean project uses Lean v4.33.1; the exact Mathlib revision is pinned in `lakefile.lean` and `lake-manifest.json`. Run `lake build` from the `lean/` directory; run `tectonic solution.tex` from the submission directory to regenerate this PDF. No auxiliary numerical computation is needed.